

A research team at Philips was able to magnetically control a swarm of screws and then superimposing more magnetic fields to individually control those screws. This new technology would allow doctors to perform targeted medications inside human bodies. Like opening cancer treating radioactive pills only at the area of the tumor while preventing exposure to the healthy parts. Another possible use could be in medical implants, where the shape of the implant could be altered based on the amount of healing done by the patient or vice versa.

Harvard University and Boston Children's Hospital have together developed a soft robot which can be wrapped around the heart to assist in its beating. Unlike existing methods to assist beating of the heart, the soft robot sleeve doesn't contact blood directly. This reduces chances of clotting, which in turn prevents patients from using blood thinners. The thin silicone sleeve is attached by pneumatic actuators to twist and compress the heart, mim-



The Leka bot

icking its beating. These actuators are powered by an external air pump. The sleeve itself is attached to the heart using a combination of a suction device, sutures, and a gel interface to reduce the friction. The sleeve can also be customised for different patients depending on specific weak areas.

Wrapping up

Even though the pop culture might have scared us about our future with robots, there is ample work going on to make them safe. Regulatory agencies are conducting strict tests and are controlling the development with guidelines. Plus, even the companies themselves have understood the fear in masses and have started taking steps to improve their technologies and also spread awareness about safe usage of robotics. As it was mentioned earlier, we're just at the gateway of this technology. There's a lot humans